

Trees May Be More Vulnerable to Climate Change Than Models Predict

By Brendan Clark

Scientists use sophisticated computer models to understand how climate change may affect the future of our planet. Among these are land surface models, which simulate processes occurring on Earth's land, including how ecosystems and vegetation respond to changing environmental conditions. These models help researchers predict how climate change may influence plant growth, forest productivity, and the amount of carbon absorbed by land ecosystems in the future, but they may be missing some important processes, according to a new study published in *Geophysical Research Letters*.

Land surface models assume that when a tree is photosynthesizing, converting sunlight into energy, it is also growing. However, recent advances in tree growth measurements have revealed that trees can continue to photosynthesize even when growth has stopped. This can occur when environmental stresses, such as insufficient water within plant tissues, prevent cells from expanding and dividing, halting growth despite continued carbon uptake through photosynthesis.

Because this decoupling between photosynthesis and growth has only recently been recognized by tree ecologists, current land surface models do not account for environmental stresses that can halt tree growth even while photosynthesis continues. Brendan Clark, a postdoctoral research associate in Cornell University's Department of Earth and Atmospheric Sciences, and his colleagues used field measurements of tree growth to develop a statistical model and compare its predictions with those from a widely used land surface model. Their results show that the land model underestimates the negative effects of increasing atmospheric dryness on tree growth, suggesting that future forest growth may be more vulnerable to climate change than current computer models indicate.

Brendan Clark, the corresponding author on the study said: "Land surface modelers may need to consider adding some kind of representation of these physiological constraints on growth to better predict climate change impacts on forests and carbon storage."

Coauthors on the study include Daniele Visoni, assistant professor in Cornell University's Department of Earth and Atmospheric Sciences; Shan Kothari, assistant professor at the University of Alberta; and Manuel Lerdau, professor at the University of Virginia.